AI-Enabled Supply Chain Management Information System

System Description, Architecture, and Multi-Stage AI Workflow

1. Introduction and System Overview

An AI-enabled Supply Chain Management Information System (AI-SCMIS) is an integrated information system that supports the planning, procurement, storage, and distribution of goods by combining operational data with artificial intelligence (AI). Instead of relying only on fixed rules and manual estimates, the system uses machine-learning models to forecast demand, detect unusual transactions, estimate supplier and delivery risks, and recommend inventory or logistics decisions. Staff remain responsible for reviewing consequential recommendations, particularly when the data is incomplete or the model's confidence is low.

This example describes a fictional distributor that supplies spare parts and equipment to several regional warehouses and customer service centres. The organization receives orders from a sales platform, purchases goods from external suppliers, stores inventory in warehouses, and dispatches deliveries through contracted carriers. Its objective is to improve product availability and delivery reliability while reducing excess stock, avoidable transport costs, and disruptions. The scenario is illustrative; a real implementation would require validation using the organization's data, processes, legal obligations, and risk appetite.

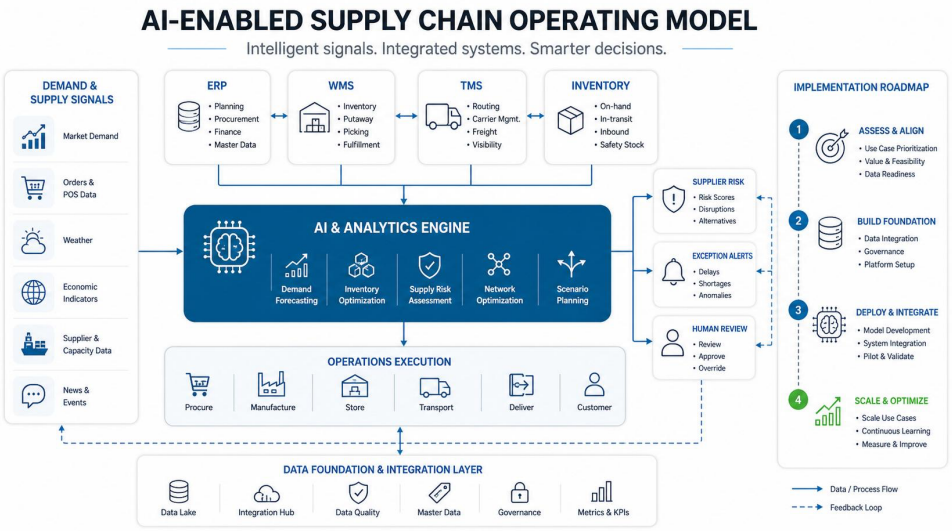
2. Business Objectives and Functional Requirements

The system has five main objectives: (1) predict product demand and seasonal variations; (2) recommend replenishment quantities and timing; (3) identify abnormal orders, stock movements, or supplier performance; (4) anticipate delays and support alternative delivery plans; and (5) provide managers with traceable, data-supported decisions. Core functions include order management, inventory tracking, supplier records, warehouse operations, shipment tracking, dashboards, notifications, and reporting.

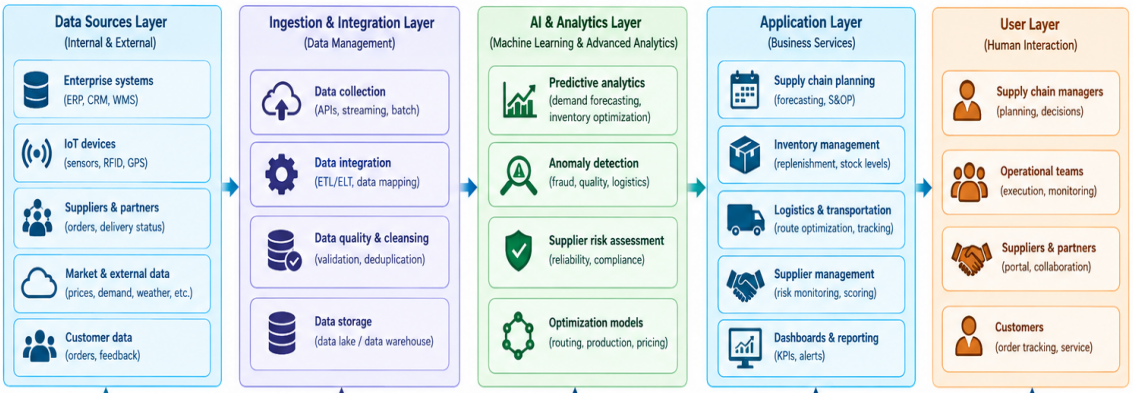
Functional requirements include consistent product and supplier identifiers, timely synchronization between sales and warehouse records, traceable model outputs, role-based access, and the ability to override AI recommendations. Non-functional requirements include availability, acceptable response time, data quality, scalability, confidentiality, integrity, auditability, and recoverability. AI predictions should be treated as decision support rather than unquestionable facts. The system must record which model version generated a recommendation and which input data and business rules were used.

3. System Architecture

The general architecture of the system is illustrated by the following figure:



The next figure resumes the functions of the above architecture in their corresponding layers:



3.1 Data sources and operational systems

The data-source layer includes customer orders, point-of-sale history, product catalogues, current stock levels, warehouse receipts, supplier lead times, purchase orders, carrier tracking events, calendar effects, and approved external indicators such as public holidays or weather forecasts. Data may arrive through application programming interfaces (APIs), scheduled imports, barcode scanners, warehouse devices, or manual entries. Each source should have an identified owner, an agreed update frequency, and defined quality checks. Personal data should be limited to what is necessary for fulfilment and customer service.

3.2 Integration and data management layer

An integration service receives and validates records, checks required fields, standardizes units and dates, and matches product and supplier identifiers. A central operational database stores current orders and stock, while a historical data store retains approved records for analytics. Data-quality services flag duplicates, missing values, implausible quantities, and inconsistent timestamps. Access to raw data is controlled, and data lineage is maintained so that analysts can trace model inputs to their source.

3.3 AI and analytics layer

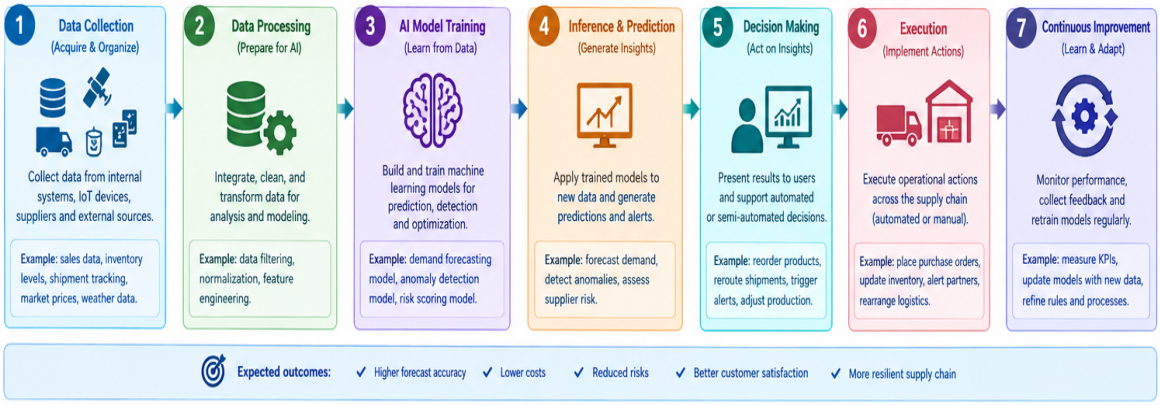
The AI layer contains separate services for demand forecasting, inventory recommendations, anomaly detection, supplier-risk estimation, and delivery-time prediction. Models are trained and evaluated on historical data that reflects relevant business conditions. A model registry records model versions, evaluation results, owners, approval status, and deployment dates. Monitoring detects changes in prediction error or input-data patterns. A model that performs poorly should be reviewed, retrained under controlled procedures, or rolled back to a validated version.

3.4 Application, user, and governance layer

Users access the system through role-specific dashboards. Warehouse staff view stock and picking tasks; procurement staff review replenishment and supplier alerts; logistics coordinators review shipment risks; managers monitor performance indicators. High-impact actions, such as placing large purchase orders or changing a critical supplier, require approval according to organizational thresholds. Logs record user activity, recommendation details, approvals, overrides, and resulting actions.

4. Multi-Stage AI Workflow

The system operates as a sequence of connected stages. Each stage produces an output that can be used by a later stage, while quality checks and human review prevent an uncertain prediction from automatically causing an inappropriate business action as shown in the following figure.



Stage 1 — Data collection and preparation

The system collects sales, order, inventory, supplier, and delivery records. It validates formats, removes duplicates where appropriate, handles missing values according to documented rules, and aligns records to consistent time periods. Data is divided into training, validation, and test sets without allowing future information to leak into past predictions. Data owners review quality exceptions before the data is approved for model use.

Stage 2 — AI-based demand forecasting

A forecasting model estimates future demand for each product and warehouse over a chosen planning horizon. Candidate approaches may include seasonal statistical models, gradient-boosted trees, or other time-series methods. Inputs can include historical sales, promotions, seasonality, lead times, and known calendar effects. The system presents a forecast and, where possible, an uncertainty range. Forecast quality is measured on held-out data and compared with a simple baseline, such as the previous-period or seasonal-naive forecast.

Stage 3 — Inventory optimization and replenishment

The forecast is combined with current stock, goods already on order, supplier lead times, safety-stock requirements, storage capacity, and budget constraints. An optimization or rules-based engine recommends what quantity to order and when. The recommendation should explain key drivers, such as predicted demand or a long supplier lead time. Procurement staff approve exceptions and high-value orders. The organization should avoid fully automatic purchasing until controls and performance have been demonstrated.

Stage 4 — Anomaly and fraud-indicator detection

A separate model or detection engine identifies unusual patterns, for example an unexpected stock reduction, repeated order changes, duplicate invoices, or an order quantity far outside the normal range. An anomaly is not proof of fraud: it is a signal for investigation. Alerts include the affected record, the reason for flagging it, and relevant historical context. Authorized staff verify the event before taking corrective action or escalating it.

Stage 5 — Supplier and delivery risk prediction

The system estimates the probability of supplier delay or late delivery using historical lead times, current shipment events, supplier performance, and approved contextual data. A high-risk prediction can prompt staff to contact the supplier, reserve alternative stock, or compare approved carriers. Recommendations should respect contractual requirements and avoid relying on variables that create unjustified or discriminatory treatment of suppliers.

Stage 6 — Human decision, execution, and feedback

The dashboard combines the forecast, stock position, anomaly alerts, and delivery risks into a proposed plan. An authorized employee accepts, modifies, or rejects the recommendation, with a reason recorded for material overrides. After orders and shipments are completed, actual demand, delivery dates, and stock movements are returned to the data platform. These outcomes support performance monitoring and future model improvement, subject to data-quality checks and controlled retraining.

5. Illustrative Data Workflow

Step	Input	AI or system activity	Output
1. Collect	Orders, sales, stock, suppliers	Validate and integrate records	Approved dataset
2. Predict	Historical demand and context	Forecast product demand	Forecast and uncertainty
3. Optimize	Forecast, stock, constraints	Calculate replenishment options	Proposed purchase plan
4. Detect	Transactions and stock changes	Flag unusual patterns	Investigative alerts
5. Anticipate	Supplier and shipment history	Estimate delay risk	Risk score and alternatives
6. Decide	Recommendations and business rules	Human review and authorization	Approved action and audit trail
7. Learn	Actual sales and outcomes	Monitor model performance	Validated improvement cycle

Example: demand forecasts indicate that a frequently used spare part will be needed in greater quantities next month. Inventory optimization detects that current stock and confirmed deliveries will not cover the forecast. Supplier-risk analysis then identifies a likely delay from the usual supplier. The system proposes an earlier order or an approved alternative supplier. A procurement officer checks the recommendation, budget, and contractual conditions before authorizing the purchase.

6. Evaluation, Benefits, and Limitations

System evaluation should measure both business performance and AI quality. Relevant indicators include forecast error (for example, MAE or WAPE), stockout rate, inventory turnover, excess inventory, on-time delivery rate, supplier-delay detection quality, false-alert rate, order-processing time, system availability, recovery performance, and security incidents. Metrics should be compared with a documented baseline and reviewed over representative periods, including seasonal peaks.

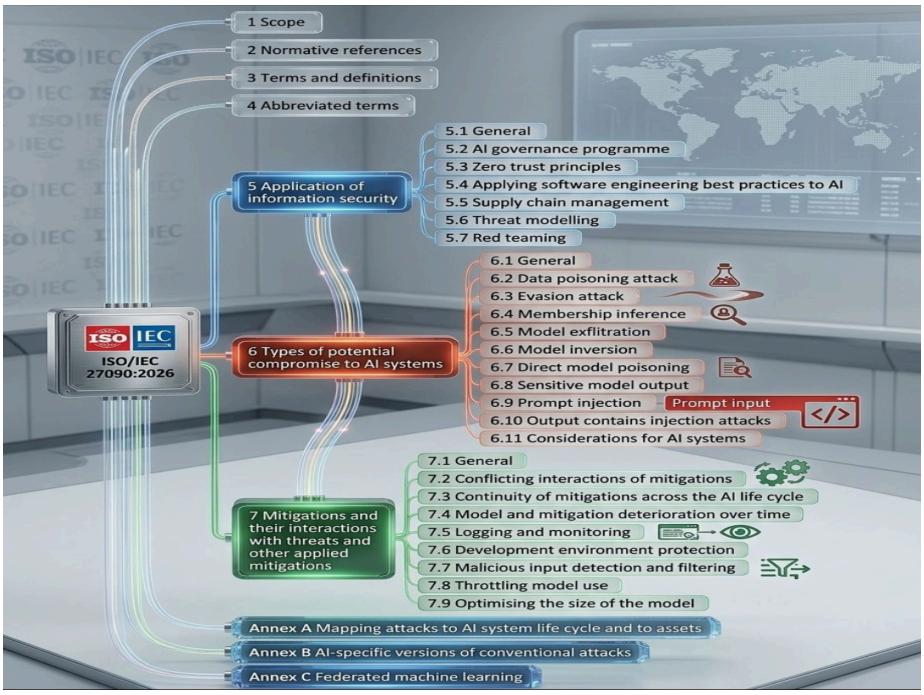
Potential benefits include improved product availability, more consistent planning, earlier detection of disruptions, reduced manual analysis, and better traceability of decisions. Limitations include poor or biased historical data, sudden market changes, model drift, integration costs, explainability challenges, and cybersecurity exposure. AI should not be presented as guaranteeing accurate forecasts or eliminating human error. A pilot should begin with a limited set of products and warehouses, compare AI recommendations with existing practice, and expand only after measured benefits and risks are reviewed.

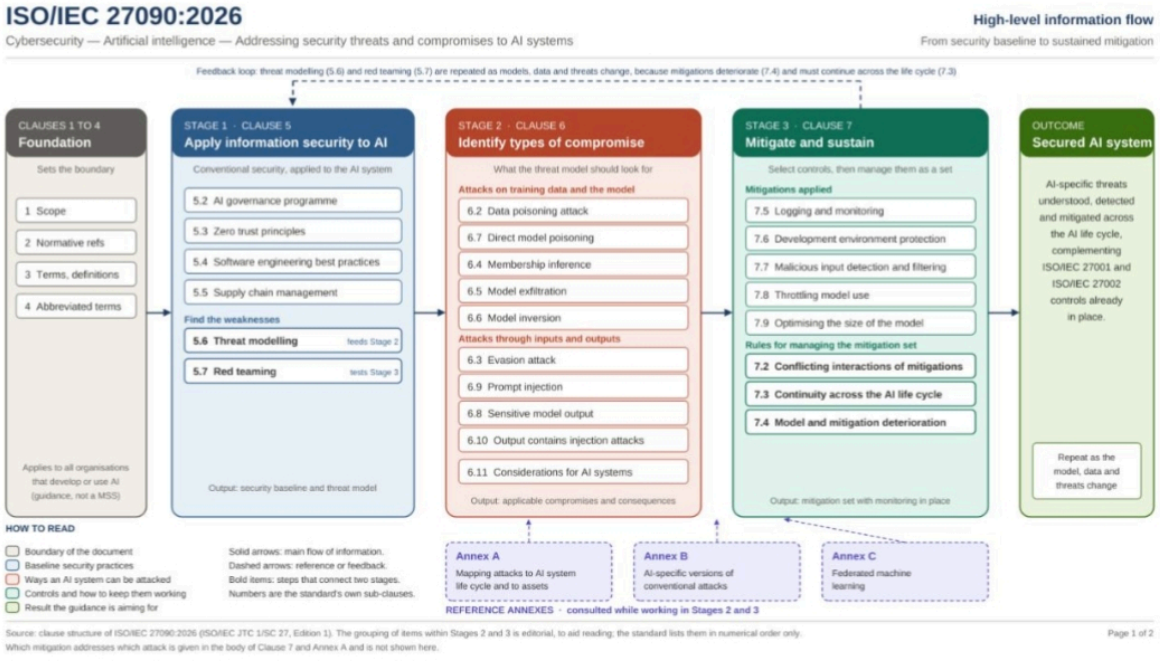
7. Conclusion

The AI-enabled supply chain information system illustrates how artificial intelligence can be applied at several connected stages: data preparation, demand forecasting, inventory optimization, anomaly detection, supplier and delivery risk prediction, and feedback-driven improvement. The system also provides a practical case study for information security management because it depends on critical data, external integrations, privileged users, AI models, and business processes that must remain available during disruptions. A sound implementation combines technical controls with governance, human oversight, risk assessment, and tested continuity procedures.

Useful information:

Since your project is oriented on AI risks in the organization, it is helpful to know about the new orientations about AI threats, especially the international norm ISO/IEC27090. The following figures illustrate the general composition of the norm (you might be interested in your project by the point 6 of the first figure – types of potential compromise to AI systems –):





A useful description is also presented in the second PDF that is communicated with this project.